Formalizing the Reduced Mahler $3/2$ Dynamical System in Lean 4: Exact 1D Slack Reduction, Constructive Coasting Survivors, Conditional Finiteness Theorems, and Certified Transcendence Barriers

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Toolchain: Lean 4.19.0 + Mathlib v4.19.0 (`114` compiled modules, `0` `sorry`s, `0` warnings, standard Lean axioms `[propext, Classical.choice, Quot.sound]`)

Abstract

Mahler's $3/2$ problem (1968) asks whether there exists any real "Z-number" $\alpha > 0$ such that the fractional part $\{(3/2)^n \alpha\}$ lies in $[0, 1/2)$ for every integer $n \geq 0$. Writing $\alpha = a_0 + \xi$ with $a_0 = \lfloor \alpha \rfloor \in \mathbb{N}$ and $\xi \in [0, 1)$, survival up to step N is equivalent to the discrete Mahler orbit $g_{n+1} = \lfloor 3g_n/2 \rfloor$ ($g_0 = a_0$) admitting a non-empty intersection of N nested archimedean companion intervals $[\mathbf{lowerBound}(a_0, n),\ \mathbf{upperBound}(a_0, n))$.

In this work, we present a 114-module, `sorry`-free formalization in Lean 4 that investigates both **constructive survival** and **obstructions to infinite orbits**:

1. **Exact 1D Dynamical Reduction:** By grouping each odd step $g \mapsto (3g+1)/2$ with its $s(k) - 1$ subsequent even steps $g \mapsto 3g/2$ via $3b_k + 1 = 2^{s(k)} u(k)$ ($\gcd(u(k), 6) = 1$), we collapse all $\binom{K}{2}$ pairwise companion-interval inequalities into a **single 1D recurrence** (`minSlack`), and prove its exact two-way equivalence with the existence of a real companion $\xi \in \mathbb{R}$. Normalizing by $2^{T(m)}$ transforms the problem into an autonomous 1D piecewise-expanding interval map $\mu_2(s, m+1) = \min\left(1,\ (3/2)^{s(m)} \mu_2(s, m) - 2\right)$ on $(0, 1]$ governed by a disjoint Lüroth–Markov partition $(0, 1] = \bigsqcup_{c=2}^{\infty} (2(2/3)^c,\ 3(2/3)^c] \cap (0, 1]$.
2. **Constructive Survivors & 113-Step Record-Holder:** We prove that any block sequence with $s(k) \geq 3$ saturates the ceiling $\mathbf{minSlack}\ s\ m = 2^{T(m)}$ and automatically admits a valid real companion. Using this sufficiency theorem, we construct closed-form seed families surviving arbitrarily many blocks prior to ceiling exit, and machine-certify a 55-bit integer (`recordHolderA0 = 29871555608238677 < 2^55`) that survives **113 steps** (58 steps past its 55-bit binary length) before dying at step 114.
3. **Unconditional Finiteness in Restricted Regimes:** We prove explicit, logarithm-free linear step cutoffs ruling out infinite orbits in four major regimes: (i) bounded-block runs ($s(k) \leq c$), (ii) periodic, eventually periodic, or low-subword-complexity block sequences ($\mathbf{blockSteps}^{(r_1)} L \leq T(r_2) + W$), (iii) quotients $u(m)$ supported on a fixed finite prime set (S-units, via Baker–Matveev), and (iv) linearly growing blocks ($s(m) \geq \delta T(m)$) or exponential slack pinches ($\Delta_m \leq 2^{(1-\delta)T(m)}$, via Ridout).
4. **Certified Diagnosis of Global Transcendence Barriers (Dead Ends 1–18):** We formally prove *why* standard elementary, polynomial-determinant, Baker–Matveev, Ridout, Schlickewei Subspace, Stewart–Bugeaud base-digit, and Padé–Hankel methods stall on a general aperiodic sequence s: each method hits an exact algebraic conservation law (rank-1 determinant factorization, linear Weil height growth

$h(u(m+1)/u(m)) \sim T(m)\log(3/2)$, exact tangency at Ridout's critical Q_m^{-2} exponent via $(1-\frac{\log 2}{\log 3})+\frac{\log 2}{\log 3}+1=2$, or a rank-2 hyperplane syzygy across shifted origins).

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1. Introduction: From Informal Proof Brainstorming to Machine-Checked Rigor

When attempting to prove Mahler's $3/2$ conjecture or related Collatz-type problems, informal proof sketches typically fall into two traps:

1. **The Probabilistic / Density Fallacy:** Treating 2-adic parity constraints and archimedean interval constraints as statistically independent, or assuming that once the step count T exceeds the bit-length $W = \lfloor \log_2 a_0 \rfloor + 1$ (the "coasting regime"), a trajectory can only survive a handful of steps.
2. **The Hidden Rank-Collapse Trap:** Constructing auxiliary linear combinations, Wronskians, or multi-origin determinants that appear to overdetermine the orbit, without noticing that the underlying 1D affine recurrence forces those combinations to be algebraically dependent.

By formalizing every proposed attack vector in Lean 4—and requiring that every failed approach be diagnosed via a machine-checked counterexample or algebraic identity—we separated genuine number-theoretic structure from mirages.

1.5 Literature Survey, Attribution of Prior Work & Novel Contributions

To place this formalization in its proper mathematical context, we carefully distinguish **prior theorems from the literature that we rediscovered or formalized** from **novel formulations, theorems, and certified barrier analyses originating in this work**.

A. Attribution to Prior Mathematical Literature

1. **Mahler (1968) — Discrete Companion Reduction, Series Formula & Two-Seed Ridout Bound [1]:**

- Kurt Mahler introduced Z-numbers and proved the foundational reduction: for each integer $g_0 = a_0 \geq 1$, there is *at most one* real companion $\xi \in [0,1)$ such that $\alpha = a_0 + \xi$ is a Z-number, governed by the discrete recurrence $g_{n+1} = \lfloor 3g_n/2 \rfloor$ and nested archimedean intervals (`MahlerDemandA.lean`, `PureIntegerPinch.lean`).
- Mahler also derived the infinite series representation of the unique candidate companion ξ_∞ (`InfiniteSeries.lean`, `TelescopingSeriesDuality.lean`) and applied **Ridout's Theorem (1957) [2]** to the difference of two hypothetical Z-numbers in the same binary interval $[2^k, 2^{k+1})$ to show that at most finitely many Z-numbers exist per interval (`mahler_two_seed_same_step_ridout_bound` and `two_seed_shared_prefix_divisibility` in `RidoutSubspaceFramework.lean`), yielding $|\{Z\text{-numbers} \leq X\}| = O(X^{0.7})$.

1. **Vijayaraghavan (1940), Pollington (1979) & Choquet (1980) — Distribution of Powers of $3/2$ and the $1/3$ Range Gap [3, 4, 5]:**

- Vijayaraghavan [3] proved that fractional parts of rational powers $(p/q)^n\alpha$ for $\alpha \in \mathbb{Q}^\times$ have infinitely many limit points and cannot stay confined to $[0, 1/2)$.
- Pollington [4] and Choquet [5] established lower bounds on the oscillation of $\{(3/2)^n\alpha\}$, leading to the critical gap constant $1/3$.

1. **Flatto, Lagarias & Pollington (1995) — The $1/3$ Range Theorem, Symbolic Dynamics & Forbidden Subwords [6]:**

- In their landmark paper [6], Flatto, Lagarias, and Pollington proved that for every real $\alpha > 0$,

$$\limsup_{n\to\infty}\{(3/2)^n\alpha\} - \liminf_{n\to\infty}\{(3/2)^n\alpha\} \geq \frac{3-2}{3} = \frac{1}{3}.$$

  This $1/3$ range gap is the continuous foundation of our $\frac{1}{3}$ **Archimedean Drain Budget** ($\sum_{k=1}^{m}(2/3)^{T(k)+1} < 1/3$ in `TelescopingSeriesDuality.lean`) and explains why $s(k) = 1$ (which consumes $(2/3)^2 = 4/9 > 1/3$ in one block) is forbidden (`EscapeBranchTwoCap.lean`).

- Flatto, Lagarias, and Pollington also developed the **symbolic dynamics of the parity sequence $g_n \bmod 2$** , characterizing admissible finite binary words via an infinite hierarchy of forbidden subwords and deriving average step-rate / topological entropy bounds. Our theorems in `GlobalStepRateBounds.lean` and `MinSlackDynamicalSystem.lean` —including $s(k) \geq 2$, the post-2 recovery requirements $s(k+1) \geq 6, s(k+2) \geq 3, s(k+3) \geq 3$, the $(2, 6, 3, 3, 2, 10, 7)$ cascade, and the $3m - 1$ step-rate bound—rediscover and machine-check these symbolic-dynamical constraints in odd-block length coordinates $s(k)$.

1. **Classical Diophantine & Transcendence Machinery (Ridout [2], Schlickewei [7], Stewart [8], Baker–Wüstholz / Matveev [9, 10], Corvaja–Zannier [11, 12], Adamczewski–Bugeaud [13], Dubickas [14]):**

- Our analyses in Modules 89–102 adapt the classical frameworks of linear forms in logarithms (Baker–Wüstholz [9], Matveev [10]), p-adic Diophantine approximation (Ridout [2], Schlickewei's p-adic Subspace Theorem [7]), simultaneous representation in bases 2 and 3 (Stewart [8]), S-unit GCD bounds (Corvaja–Zannier [11, 12]), and subword-complexity transcendence criteria (Adamczewski–Bugeaud [13]).

B. Novel Contributions Originating in This Work (and Comparison with Prior Lean Formalizations)

1. **Relationship to Prior Lean Repositories on Mahler's Work:**

- A direct search of all public Lean repositories on GitHub shows that **Google DeepMind's `formal-conjectures` project [15]** (`FormalConjectures/Wikipedia/Mahler32.lean` , `FormalConjectures/Books/BugeaudDistributionModuloOne/Problem10_9.lean` , and `FormalConjecturesForMathlib/NumberTheory/DiophantineApproximation/ZNumber.lean` , 2025–2026) previously formalized the *definition* `IsZNumber` , a 5-line sanity check `¬ IsZNumber 1` , and the *statements* (with `sorry` placeholders) of Mahler's conjecture, the consequence of $\Omega(3/2) > 1/2$, the Flatto–Lagarias–Pollington bound $\Omega(p/q) \geq 1/p$, Lebesgue measure zero, and Hausdorff dimension < 1.
- Other Lean 4 repositories bearing Mahler's name address distinct theorems of Kurt Mahler: `giuliocaflisch/mahler` (Mahler's theorem on p-adic continuous functions),

`DiegoMarquesMath/MahlerLean` (Mahler's classification of Liouville numbers), and `baobingzhang/jsp-000025-lehmer-schinzel-lean` (polynomial Mahler measure).

- Thus, while `google-deepmind/formal-conjectures` [15] formalized the top-level conjecture statements for Z-numbers, our 114-module `sorry`-free library is the **first proved Lean 4 development of the discrete Mahler $3/2$ dynamical system**, the companion-interval reduction, the Flatto–Lagarias–Pollington symbolic forbidden-subword constraints in odd-block coordinates, and the transcendence/Diophantine barrier analyses.

1. **The Reduced Coprime Quotient System & Exact 1D `minSlack` Dynamical Collapse (`CoprimeRemainderSlack.lean`, `MinSlackDynamicalSystem.lean`, `ConverseCompanionReconstruction.lean`, `CoupledFractionalPartDynamics.lean`):**

- Formulating the dynamics in terms of the coprime odd quotients $2^{s(k+1)}u(k+1) = 3^{s(k)}u(k)+1$ ($\gcd(u(k),6)=1$), the 3-adic block-boundary factorization $b_{k+1} = 3^{s(k)-1}u(k)$, and the polynomial $R(s,m)$.
- Collapsing all $\binom{K}{2}$ shifted archimedean companion inequalities into the **single 1D recurrence** `minSlack s (m+1) = min (2^(T(m+1))) (3^(s m) * minSlack s m - 2^(T(m+1)+1))` and its autonomous 1D interval map $\mu_2(s,m+1) = \min\left(1,\ (3/2)^{s(m)}\mu_2(s,m) - 2\right)$ on the disjoint Lüroth–Markov partition $(0,1] = \bigsqcup_{c=2}^{\infty}(2(2/3)^c, 3(2/3)^c] \cap (0,1]$, together with the two-way **Converse Companion Reconstruction Theorem** (`(∃ ξ, ValidCompanionUpTo a_0 ξ (blockSteps s K)) ↔ 0 < minSlack s (K - 1)`).

1. **Additive Separation of Shifted Slacks & the Three-Origin Subspace Syzygy (`MultiOriginSubspaceSyzygy.lean`):**

- Proving that the two-parameter triangle of shifted slacks separates additively as $\frac{2^{T(r)}\Delta_{m-r}^{(r)}}{3^{T(m)}} = \frac{\Delta_m^{(0)}}{3^{T(m)}} - \frac{\Delta_r^{(0)} - 2^{T(r)}}{3^{T(r)}}$, eliminating `shiftSeq` completely from `0 < minSlack s m`, and proving the exact **Three-Origin Schlickewei Subspace Hyperplane Syzygy** (Dead End 15) showing that any ≥ 3 shifted origins lie in the 2D plane $\mathbf{span}\{\Delta_m^{(0)}, 3^{T(m)}\}$.

1. **Carryless Base-3 Frozen-Prefix Equivalence & Dual Remainder Complementarity (`SimultaneousBaseDigitRigidity.lean`):**

- Proving $0 < \Delta_m \iff \lfloor 2^{T(m+1)+1}u(m)/3^{T(m)} \rfloor = 6a_0+2$ (and its multi-origin companion equivalence) together with the exact dual remainder law $(P_m \bmod 3^{T(m)}) + (Q_m \bmod 2^{T(m)}) = 3^{T(m)}$.

1. **Exact Three-Way Multi-Step GCD Identities & Two-Sided CRT Quotient Floors (`MultiStepQuotientGCD.lean`, `ThreeAdicAndSeedGCDLocks.lean`, `Mod2304CombinedSieve.lean`, `TwoSidedCRTAndBackwardObstructions.lean`, `Mod216TwoSidedQuotientFloor.lean`):**

- Proving the exact GCD equalities $\gcd(u(r),u(r+d)) = \gcd(u(r),R_d^{(r)}) = \gcd(u(r+d),R_d^{(r)})$, $\gcd(\Delta_r,\Delta_{r+d}) = \gcd(\Delta_r,R_d^{(r)})$ (with $\gcd(\Delta_r,\Delta_{r+1})=1$), and $\gcd(u(m),\Delta_m) = \gcd(u(m),2a_0+1) = \gcd(\Delta_m,2a_0+1)$.
- Proving the closed-form 3-adic residue lock $u(k+1) \bmod 3^{s(k)} = ((3^{s(k)}+1)/2)^{s(k+1)} \bmod 3^{s(k)}$ and combining backward 3-adic congruences ($\bmod\ 3,9,27$) with forward 2-adic congruences (

$\bmod 4, 8, 64, 256$) to prove $u(k) \neq 1$ ($k \geq 1$), $1735 \leq u(k)$ at $s(k) = 2$, $65 \leq u(k)$ across $2 \leq s(k) \leq 6$, and $103 \leq u(k)$ on ceiling-saturated runs.

1. **Exact $2^{T(m)}$-Cancellation in the Single-Origin Ridout Product & Explicit Repeated-Subword Cutoffs (`RidoutSubspaceFramework.lean`, `RepeatedSubwordComplexity.lean`):**

- Proving the single-origin cancellation $2^{s(m)+D+1} u(m) \Delta_m < 3^{T(m)+1}(2a_0 + 1)$ and the exponent identity $(1 - \frac{\log 2}{\log 3}) + \frac{\log 2}{\log 3} + 1 = 2$ (Dead End 14).
- Proving the parity-enhanced common multiplier $u(r_2) - u(r_1) = 2^{S_{\text{tail}}+1} K_0$, $u(r_2 + d) - u(r_1 + d) = 2 \cdot 3^S K_0$ across any repeated subword and its logarithm-free linear cutoff $\text{blockSteps}^{(r_1)} L \leq T(r_2) + W$.

1. **Discovery & Machine Certification of the 55-Bit, 113-Step Coasting Record-Holder (`MahlerGrandSynthesis.lean`):**

- Using our ceiling-saturation sufficiency theorem (`exists_validCompanionUpTo_of_three_le`) to construct a targeted congruence search that uncovered `recordHolderA0 = 29871555608238677 < 2^55`, surviving 113 steps (58 steps past its 55-bit binary length) and dying at step 114.

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2. The Odd-Block Reduction & Dual 2-Adic / 3-Adic Recurrence

Let `step x = (3 * x + 1) / 2` if `x` is odd and `3 * x / 2` if `x` is even, and let `orbit a_0 t` be the t-th iterate starting from $a_0 \in \mathbb{N}$.

A sequence of block lengths $s : \mathbb{N} \to \mathbb{N}$ and odd quotients $u : \mathbb{N} \to \mathbb{N}$ forms a **`ValidBlockChain a_0 s u K`** (`OddBlockReduction.lean`) if for every $k < K$:

$$1 \leq s(k), \qquad u(k) \bmod 2 = 1, \qquad 3 \cdot b_k + 1 = 2^{s(k)} u(k),$$

where $b_k := \text{orbit } a_0 \ (T(k))$ and $T(k) := \text{blockSteps } s \ k = \sum_{i=0}^{k-1} s(i)$.

Fundamental Identities Along Any Valid Block Chain

- **One-Step Coprime Quotient Recurrence (`CoprimeRemainderSlack.lean`):**

$$2^{s(k+1)} u(k+1) = 3^{s(k)} u(k) + 1 \qquad (k + 1 < K).$$

- **Exact 3-Adic Factorization of Block-Boundary Orbit Values (`ThreeAdicAndSeedGCDLocks.lean`):**

$$b_{k+1} = \text{orbit } a_0 \ (T(k+1)) = 3^{s(k)-1} \cdot u(k), \qquad \gcd(u(k), 6) = 1.$$

  Thus at every block boundary b_{k+1}, its **3-adic valuation** is $s(k) - 1$ (encoding the *preceding* block length) with 3-free part $u(k)$, while the **2-adic valuation of $3b_{k+1} + 1$** is $s(k+1)$ (encoding the *upcoming* block length) with 2-free part $u(k+1)$!

- **Multi-Step Coprime Remainder & Intrinsic Slack (`CoprimeRemainderSlack.lean`, `AperiodicSlackCocycle.lean`):** Defining $R(s, 0) = 0$ and $R(s, m+1) = 3^{s(m)} R(s, m) + 2^{\text{tailSteps } s \ m}$ (where $\text{tailSteps } s \ m = \sum_{i=1}^{m} s(i)$), the m-step quotient satisfies:

$$2^{\text{tailSteps } s\ m} u(m) = 3^{T(m)} u(0) + R(s,m).$$

Eliminating $u(0)$ via $2^{s(0)}u(0) = 3a_0 + 1$ yields the **Orbit Slack Identity**:

$$\Delta_m := \text{integerSlack } s\ m := 3^{T(m)} - 2^{s(0)+1} R(s,m) = 3^{T(m)+1}(2a_0+1) - 2^{T(m+1)+1} u(m).$$

Notice that the middle expression $3^{T(m)} - 2^{s(0)+1}R(s,m)$ depends **solely on the block sequence s** and is completely independent of a_0 and u!

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3. Exact 1D Dynamical Reduction & Converse Companion Reconstruction

At first glance, ensuring that all K archimedean companion intervals $[\text{lowerBound}(a_0, T(j)),\ \text{upperBound}(a_0, T(m+1)))$ overlap requires checking $\binom{K}{2}$ pairwise inequalities:

$$\forall\, 0 \le j \le m < K, \qquad 0 < \Delta_{m-j}^{(j)} := \text{integerSlack } (\text{shiftSeq } s\ j)\ (m-j).$$

We proved four equivalent ways to collapse this $\binom{K}{2}$ triangular system into a **1D dynamical condition**:

1. **The 1D `minSlack` Dynamical System (`MinSlackDynamicalSystem.lean`, `ConverseCompanionReconstruction.lean`):** Define:

$$\text{minSlack } s\ 0 = 1, \qquad \text{minSlack } s\ (m+1) = \min\Big(2^{T(m+1)},\ 3^{s(m)} \cdot \text{minSlack } s\ m - 2^{T(m+1)+1}\Big).$$

Then for any `ValidBlockChain a_0 s u K` ($K \ge 1$), a real companion $\xi \in \mathbb{R}$ surviving through step $T(K)$ exists **if and only if** `0 < minSlack s (K - 1)`:

$$\big(\exists \xi : \mathbb{R},\ \text{ValidCompanionUpTo } a_0\ \xi\ (T(K))\big) \iff 0 < \text{minSlack } s\ (K-1).$$

1. **Shift-Free Unshifted Ceiling Comparison (`MultiOriginSubspaceSyzygy.lean`):** Using the **Shifted Slack Affine Collapse Identity**

$$2^{T(r)}\Delta_d^{(r)} = \Delta_{r+d}^{(0)} + 3^{S_{r,d}}\big(2^{T(r)} - \Delta_r^{(0)}\big),$$

every `shiftSeq` can be eliminated:

$$0 < \text{minSlack } s\ m \iff \forall j \le m, \quad 3^{T(m)}\big(\Delta_j^{(0)} - 2^{T(j)}\big) < 3^{T(j)}\Delta_m^{(0)}.$$

1. **Carryless Base-3 Shift-and-Add Automaton (`SimultaneousBaseDigitRigidity.lean`):** The scaled quotients $P_m := 2^{T(m+1)+1}u(m)$ satisfy the base-3 shift-and-add recurrence $P_{m+1} = 3^{s(m)}P_m + 2^{T(m+1)+1}$, and unconditionally $P_m = 3^{T(m)}(6a_0+2) + 2^{s(0)+1}R(s,m)$. Thus:

$$0 < \Delta_m^{(0)} \iff \left\lfloor \frac{2^{T(m+1)+1}u(m)}{3^{T(m)}} \right\rfloor = 6a_0 + 2 \quad (\text{no base-3 carry across } 3^{T(m)}!).$$

1. **Autonomous 1D Piecewise-Expanding Map & Lüroth–Markov Partition (`CoupledFractionalPartDynamics.lean`):** Normalizing by $2^{T(m)}$ via $\mu_2(s,m) := \text{minSlack } s\ m / 2^{T(m)}$ removes the explicit time dependence $T(m)$ completely:

$$\mu_2(s,0) = 1, \qquad \mu_2(s,m+1) = \min\left(1,\ \left(\frac{3}{2}\right)^{s(m)} \mu_2(s,m) - 2\right).$$

Whenever a transition stays strictly inside the sub-ceiling corridor $(0,1]$, $\theta_2(s,m) = \Delta_m/2^{T(m)}$ must lie in the partition cell $I_{s(m)} = (2(2/3)^{s(m)},\ 3(2/3)^{s(m)}]$. Because $3(2/3)^{c+1} = 2(2/3)^c$, the cells $I_c \cap (0,1]$ ($c = 2,3,4,\ldots$) form a **pairwise-disjoint partition of** $(0,1]$.

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4. Ceiling Saturation & the 113-Step 55-Bit Record-Holder

Whenever $s(k) \geq 3$ for all $k < m$, `minSlack` remains pinned at its ceiling (`CeilingSaturationSynthesis.lean`):

$$\left(\forall k < m,\ 3 \leq s(k)\right) \implies \mathbf{minSlack}\ s\ m = 2^{T(m)} > 0.$$

Consequently, **every** block chain with $s(k) \geq 3$ automatically satisfies all archimedean companion constraints (`exists_validCompanionUpTo_of_three_le`).

- **Arbitrarily Long Closed-Form Survivors (`ExplicitHorizonCutoffs.lean`):** The closed-form integer family

$$a_0(r) = \frac{92 \cdot 2^{12r} - 27}{65}$$

executes r consecutive $(3,3,3,3)$ cycles ($12r$ steps) with $s(k) = 3$, surviving arbitrarily many steps before its first $s = 2$ block.

- **The 113-Step Coasting Record-Holder (`MahlerGrandSynthesis.lean`):** Searching the ceiling-saturated congruence tree uncovered the 55-bit seed:

$$\mathbf{recordHolderA0} = 29{,}871{,}555{,}608{,}238{,}677 < 2^{55}.$$

In `recordHolderA0_exact_113_survival_and_114_death`, Lean machine-checks that `recordHolderA0` executes 31 blocks with $s(k) \geq 3$ through $T(31) = 109$, hits its first $s(31) = 2$ at step 109, takes $s(32) = 1$ at step 111, **survives 113 steps** (58 steps past its 55-bit binary length!), and dies at step 114.

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5. Unconditional Finiteness Theorems & Two-Sided CRT Sieves

While ceiling-saturated chains can be constructed up to $T(m) \approx W$ for $a_0 < 2^W$, once $T(m) > W$ (the **coasting regime**), we proved unconditional finiteness across the following regimes:

1. **Global Step-Rate & Even-Step Domination (`GlobalStepRateBounds.lean`, `ThreeAdicValuationConservation.lean`):** Every surviving orbit satisfies $s(k) \geq 2$, $s(k) = 2 \implies s(k+1) \geq 6, s(k+2) \geq 3, s(k+3) \geq 3$, and the global step-rate bound:

$$3m \leq T(m) + 1 \iff 2m \leq \mathbf{evenStepSum}\ s\ m + 1.$$

Moreover, the product of the first m block-boundary orbit values has exact 3-adic valuation $T(m) - m$:

$$\prod_{k=0}^{m-1} b_{k+1} = 3^{T(m)-m} \prod_{k=0}^{m-1} u(k), \qquad \gcd\left(\prod_{k=0}^{m-1} u(k),\ 6\right) = 1.$$

1. **Time-Independent Bounded-Block Escape Ceilings (`AperiodicBoundedBlockSqueeze.lean`, `TimeIndependentDeficiencySqueeze.lean`):** During any run of blocks bounded by $s(k+i) \le c$ after $s(k-1) = 2$, the $2^{T(k)}$ factor cancels identically from `minSlackDeficiency`, bounding the run to at most **6 steps** for $c = 3$ and **4 blocks** for $c = 5$ regardless of how large $T(k)$ is!

1. **Repeated-Subword Complexity Cutoff (`RepeatedSubwordComplexity.lean`):** Whenever an L-block subword repeats at two origins $r_1 < r_2$ ($\forall i < L,\ s(r_1+i) = s(r_2+i)$), subtracting the multi-step quotient equations cancels the unknown remainder R_{win} identically and yields a common positive integer $K_0 \ge 1$ such that:

$$u(r_2) - u(r_1) = 2^{\text{tailSteps}^{(r_1)}(L-1)+1} K_0, \qquad u(r_2+L-1) - u(r_1+L-1) = 2 \cdot 3^{\text{blockSteps}^{(r_1)}(L-1)} K_0.$$

   For $a_0 < 2^W$, this forces the explicit logarithm-free linear cutoff:

$$\text{blockSteps}\ (\text{shiftSeq}\ s\ r_1)\ L \le T(r_2) + W, \qquad 3L \le T(r_2) + W + 2,$$

   ruling out all periodic, eventually periodic, and low-complexity block sequences.

1. **Two-Sided Chinese Remainder Sieves & Quotient Floors (`Mod256CompleteSieve.lean`, `Mod2304CombinedSieve.lean`, `TwoSidedCRTAndBackwardObstructions.lean`, `Mod216TwoSidedQuotientFloor.lean`):**

- **Mod-2304 (256×9) Two-Block Sieve:** Eliminates **2283** of **2304** residue classes of $(3^{s(k)}u(k)+1) \bmod 2304$ (**99.09%**), locking the post-**2** recovery branch to the single class $Y = 28 \pmod{2304}$.
- **Unconditional Impossibility of $u(k) = 1$ ($k \ge 1$):** Looking one step backward ($k-1 \to k$) mod **3** forces $u(k) = 1 \implies s(k)$ even, and looking one step forward ($k \to k+1$) mod **4** forces $3^{s(k)} + 1 \equiv 2 \pmod 4 \implies s(k+1) = 1$ (immediate death).
- **Mod-54, Mod-216 & Mod-1728 Quotient Floors:** Combining backward Mod-27 locks ($3 \le s(k-1)$) with forward 2-adic locks forces $1735 \le u(k)$ at $s(k) = 2$, $65 \le u(k)$ across $2 \le s(k) \le 6$, and $103 \le u(k)$ on ceiling-saturated runs ($3 \le s(k) \le 6$ with $3 \le s(k+1)$).

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6. Certified Transcendence & Diophantine Barriers (Why the General Case Remains Open)

A central contribution of this formalization is proving **exact algebraic conservation laws (Dead Ends 1–18)** that explain why every standard transcendence and Diophantine method falls just short on a general aperiodic sequence s:

Barrier 1: Why Infinite Solutions Exist in $\mathbb{Z}_2 \times \mathbb{R}$ (`TwoAdicCauchyConvergence.lean`, `MahlerSeriesHankelFramework.lean`)

The single power series $\Sigma_m(s) = \sum_{k=1}^{m} (2/3)^{T(k)+1} = \frac{2^{s(0)+1} R(s,m)}{3^{T(m)+1}}$ converges in **both** $\mathbb{R}$ (since $|2/3|_\infty = 2/3 < 1$) and $\mathbb{Q}_2$ (since $|2/3|_2 = 1/2 < 1$). Along any valid block chain:

$$\Sigma_m(s) + \frac{6a_0+2}{3} = \frac{2^{T(m+1)+1}u(m)}{3^{T(m)+1}}.$$

In $\mathbb{Z}_2 \times \mathbb{R}$, **infinite solutions genuinely exist** for every sequence s with $\mathbf{minSlack}\ s\ m > 0$: there is always a unique 2-adic integer $\alpha_2 \in \mathbb{Z}_2$ and real companion $\xi_\infty \in [0,1)$ satisfying all congruences and archimedean bounds forever. A true integer counterexample $a_0 \in \mathbb{N}$ is simply a sequence s for which the 2-adic limit $\Sigma_\infty^{(\mathbb{Q}_2)}(s) = -\frac{6a_0+2}{3}$ happens to be a negative rational number with denominator 3.

Barrier 2: Rank-1 Determinant & Padé–Hankel Valuation Collapse (Dead Ends 12 & 18, `DeterminantFactorization.lean`, `MahlerSeriesHankelFramework.lean`)

When one forms the 2×2 determinant between steps r and $r+d$ to eliminate $2a_0+1$ (either from the archimedean slacks Δ_k or from the 2-adic numerators $P_k = 2^{T(k+1)+1}u(k)$), the determinant factors identically as:

$$3^{T(r+d)}\Delta_r - 3^{T(r)}\Delta_{r+d} = 3^{T(r)} \cdot 2^{T(r+1)+1} \cdot R(\mathrm{shiftSeq}\ s\ r,\ d),$$

$$3^{T(r)+1}P_{r+d} - 3^{T(r+d)+1}P_r = 3^{T(r)+1} \cdot 2^{T(r+1)+1} \cdot R(\mathrm{shiftSeq}\ s\ r,\ d).$$

Because $\gcd(R_d^{(r)}, 6) = 1$, the 2-adic valuation of the determinant drops from $T(r+d+1)+1$ all the way back to $T(r+1)+1$, preventing Padé approximation from gaining a 2-adic advantage over a single step.

Barrier 3: Baker–Matveev Linear Forms in Logarithms & Multi-Step GCDs (Dead Ends 13 & 17, `BakerMatveevFramework.lean`, `MultiStepQuotientGCD.lean`)

In `BakerMatveevFramework.lean`, we proved the exponentially small 3-logarithm linear form bounds:

$$0 < s(m+1)\log 2 - s(m)\log 3 + \log\left(\frac{u(m+1)}{u(m)}\right) < \frac{1}{3a_0+1}\left(\frac{2}{3}\right)^{T(m+1)},$$

$$0 < (T(m)+1)\log 3 - (T(m+1)+1)\log 2 + \log\left(\frac{2a_0+1}{u(m)}\right) \le \frac{1}{6a_0+2}\left(\frac{2}{3}\right)^{T(m)}.$$

When $u(m)$ is supported on a fixed finite set of primes (S-units), all algebraic bases are fixed and Matveev's lower bound $\log|\Lambda| \ge -C(S,a_0)\log T(m)$ beats $-T(m)\log(3/2)$, **unconditionally proving finiteness**.

However, in the general case:

- Consecutive quotients are always coprime ($\gcd(u(m), u(m+1)) = 1$, `consecutive_u_coprime`).
- Across d steps (`multi_step_u_gcd_eq_gcd_coprimeRemainder`), $\gcd(u(r), u(r+d)) = \gcd(u(r), R_d^{(r)}) < 3^{S_{r,d}}/2^{s(r)+1}$.
- Even after canceling $\gcd(u(r), u(r+d))$, the reduced denominator satisfies $2^{T(r)}3^{S_{r,d}}\big(u(r)/\gcd(u(r),u(r+d))\big) > 3^{T(r)}(3a_0+1)$ (`multi_step_reduced_denominator_exponential_lower_bound`). Thus the logarithmic Weil height of the variable rational term grows linearly as $h \sim T(m)\log(3/2)$, causing Matveev's lower bound $-C_3 \cdot T(m)\log T(m)$ to lose to $-T(m)\log(3/2)$ by a factor of $\log T(m)$.

Barrier 4: Ridout's Theorem & the Critical Q_m^{-2} Exponent Identity (Dead End 14, `RidoutSubspaceFramework.lean`)

In the orbit equation $Q_m - P_m = \Delta_m$ with $Q_m = 3^{T(m)+1}(2a_0+1)$ and $P_m = 2^{T(m+1)+1}u(m)$, the $\{2,3\}$-free part of P_m is $u(m) \sim (3/2)^{T(m)}$ (with $2^{T(m)}$ in the denominator), while in any sub-ceiling phase $\Delta_m \le 2^{T(m)}$ (with $2^{T(m)}$ in the numerator). Multiplying $u(m)$ and Δ_m **cancels $2^{T(m)}$ identically** (`ridout_non_s_part_cancellation`):

$$2^{s(m)+D+1} \cdot u(m) \cdot \Delta_m < 3^{T(m)+1}(2a_0+1) = Q_m.$$

Consequently, the 3-place $\{\infty, 2, 3\}$ Ridout product satisfies (`ridout_three_place_quadratic_bound`):

$$\left(1 - \frac{P_m}{Q_m}\right) \cdot \frac{u(m)}{P_m} \cdot \frac{2a_0+1}{Q_m} < \frac{(6a_0+3)(2a_0+1)}{2^{s(m)+1}(6a_0+2)} \cdot \frac{1}{Q_m^2}.$$

Because the three place exponents sum to **exact 2**:

$$\underbrace{\left(1 - \frac{\log 2}{\log 3}\right)}_{|\cdot|_\infty} + \underbrace{\left(\frac{\log 2}{\log 3}\right)}_{|\cdot|_2} + \underbrace{1}_{|\cdot|_3} = 2,$$

the general bounded-block orbit sits **right on Ridout's critical Q_m^{-2} boundary**, crossing below $Q_m^{-2-\varepsilon}$ only when $s(m) \ge \delta T(m)$ or $\Delta_m \le 2^{(1-\delta)T(m)}$.

Barrier 5: Multi-Origin Schlickewei Subspace Hyperplane Syzygy (Dead End 15, `MultiOriginSubspaceSyzygy.lean`)

Attempting to break the single-origin Q_m^{-2} tangency by combining $k \ge 3$ shifted origins $(0, r_1, r_2, \ldots)$ in Schlickewei's higher-dimensional p-adic Subspace Theorem fails because every scaled shifted slack $2^{T(r)}\Delta_{m-r}^{(r)}$ lies in the **same 2-dimensional plane $\operatorname{span}\{\Delta_m^{(0)}, 3^{T(m)}\}$** as $m \to \infty$. Any three origins $(0, r_1, r_2)$ satisfy the exact m-independent hyperplane syzygy (`three_origin_subspace_syzygy`):

$$D_0(r_2) \cdot 3^{T(r_1)}\left(2^{T(r_1)}\Delta_{m-r_1}^{(r_1)} - \Delta_m^{(0)}\right) = D_0(r_1) \cdot 3^{T(r_2)}\left(2^{T(r_2)}\Delta_{m-r_2}^{(r_2)} - \Delta_m^{(0)}\right),$$

landing identically inside the degenerate exceptional subspace of Schlickewei's theorem.

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7. Architecture of the 114 Lean 4 Modules & Verification

Every theorem in this repository is machine-checked in Lean 4.19.0 (`mahler_demand_a/`) without a single `sorry`. Each core module `Foo.lean` is paired with a verification module `FooChecks.lean` that runs `#check` and `#print axioms` on every exported theorem.

Key Module Groups

- **Foundations & Odd-Block Reduction (Modules 1–36):** `MahlerDemandA.lean` through `TwoSidedSqueeze.lean` — basic Mahler step/orbit properties, companion intervals, odd-to-odd block reduction, `coprimeRemainder`, `integerSlack`, and initial two-sided squeezes.

- **1D Dynamical Collapse & Converse Reconstruction (Modules 37–52):** `SharpenedSqueezes.lean`, `AperiodicSlackCocycle.lean`, `MinSlackDynamicalSystem.lean`, `GlobalStepRateBounds.lean`, `BaseTwoDivModLock.lean`, `CeilingSaturationSynthesis.lean`, `TelescopingSeriesDuality.lean`, `ConverseCompanionReconstruction.lean`.
- **2-Adic LTE, Cauchy Convergence & Time-Independent Bounds (Modules 53–80):** `ExplicitHorizonCutoffs.lean`, `CeilingDeficitAmplification.lean`, `UniversalTwoVisitSeparation.lean`, `MonotonicStepSumSeparation.lean`, `CeilingSaturatedModTraps.lean`, `TwoAdicLiftingExponent.lean`, `FullStepSumValuation.lean`, `CoprimeRemainderAdditionLaw.lean`, `TwoAdicCauchyConvergence.lean`, `AllHistoryQuotientRecovery.lean`, `ParametricRemainderCorridor.lean`, `AperiodicBoundedBlockSqueeze.lean`, `TimeIndependentDeficiencySqueeze.lean`, `SlackDeadEndsAndWitnesses.lean`.
- **Record-Holder, Sieves & Transcendence Frameworks (Modules 81–114):**
- `MahlerGrandSynthesis.lean`: Equivalence `validBlockChain_iff_coprime_recurrence` and 113-step survival / 114-step death of `recordHolderA0 = 29871555608238677`.
- `Mod256CompleteSieve.lean`, `CoprimeUnitRigidity.lean`, `DeterminantFactorization.lean`: Mod-256 two-block sieve, unit-sum rigidity, and rank-1 determinant collapse (Dead End 12).
- `BakerMatveevFramework.lean`: Consecutive coprimality, 3-logarithm `Real.log` squeezes, and Weil height growth (Dead End 13).
- `RidoutSubspaceFramework.lean`: Intrinsic $\gcd(\Delta_m, 6) = 1$, $2^{T(m)}$-cancellation law $2^{s(m)+1} u(m) \Delta_m < Q_m$, and Q_m^{-2} Ridout bound (Dead End 14).
- `MultiOriginSubspaceSyzygy.lean`: Additive separation of variables, shift-free `minSlack` equivalence, and 3-origin Subspace syzygy (Dead End 15).
- `RepeatedSubwordComplexity.lean`: Strict quotient aperiodicity $u(r_1) < u(r_2)$, parity-enhanced common multiplier, and linear subword-complexity cutoffs.
- `SimultaneousBaseDigitRigidity.lean`: Carryless Base-3 prefix equivalence and dual Base-2 / Base-3 remainder complementarity (Dead End 16).
- `MultiStepQuotientGCD.lean`: Exact three-way multi-step GCD equality $\gcd(u(r), u(r+d)) = \gcd(u(r), R_d^{(r)})$ and reduced-denominator exponential growth (Dead End 17).
- `MahlerSeriesHankelFramework.lean`: Power-series functional equation and Padé–Hankel determinant factorization (Dead End 18).
- `ThreeAdicAndSeedGCDLocks.lean`, `CoupledFractionalPartDynamics.lean`, `Mod2304CombinedSieve.lean`, `TwoSidedCRTAndBackwardObstructions.lean`, `ThreeAdicValuationConservation.lean`, `Mod216TwoSidedQuotientFloor.lean`: Seed/slack GCD locks, 3-adic orbit factorization $b_{k+1} = 3^{s(k)-1} u(k)$, autonomous μ_2 interval map, disjoint Lüroth–Markov partition, Mod-2304 / Mod-1728 / Mod-216 sieves, $u(k) \neq 1$ ($k \geq 1$), and 3-adic valuation conservation.

Reproducing the Verification Build

```
cd mahler_demand_a
lake build
lake env lean -DwarningAsError=true ThreeAdicValuationConservationChecks.lean
lake env lean -DwarningAsError=true Mod216TwoSidedQuotientFloorChecks.lean
```

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